

Chapter 7: Attitude Representation & Attitude Dynamics – week 1

Dr. Alexander Wittig

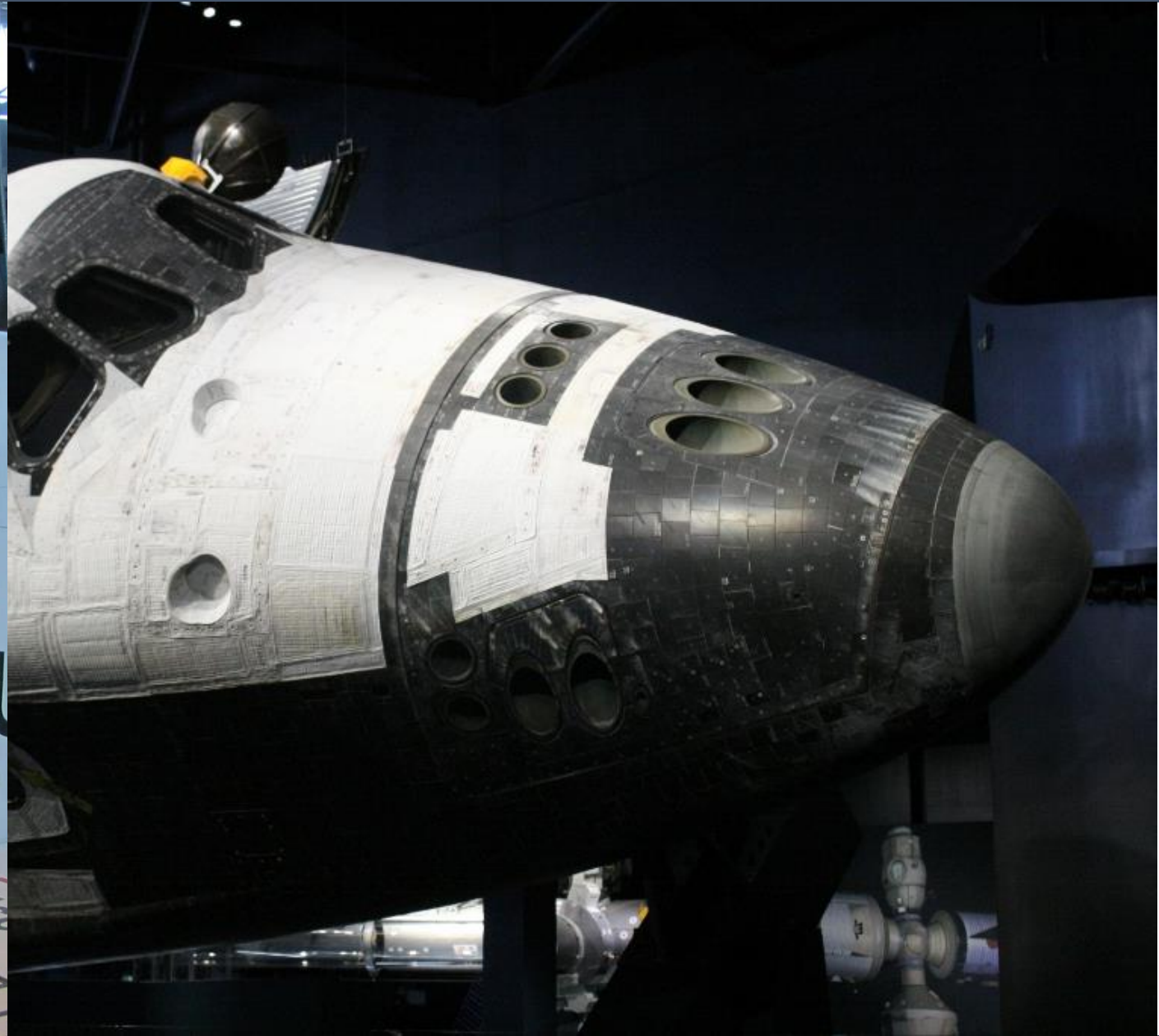
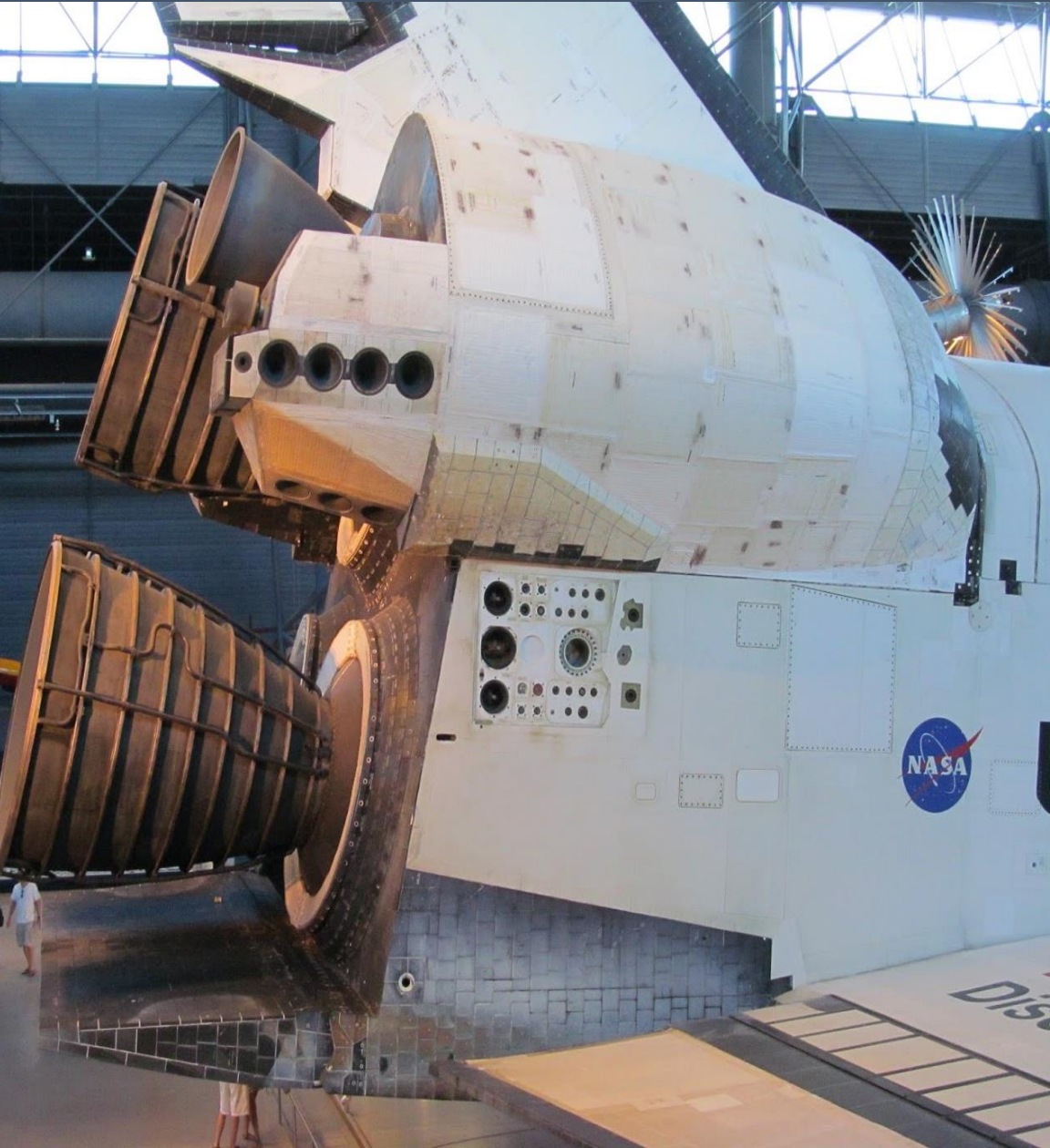
- Any questions on previous weeks content?

- **Introduction**
- **Chapter 1: In 20.000 km, turn right. Orbital Manoeuvres**
 - Coplanar Manoeuvres
 - Non-Coplanar Transfers
- **Chapter 2: That's mildly disturbing Orbital Perturbations & Propagation**
 - Non-Uniform Gravity Field
 - Third-Body Perturbation
 - Atmospheric Drag
 - Solar Radiation Pressure
 - Orbit Propagation Methods
- **Chapter 3: Is it a plane? Is it a bird? Orbit Observation & Determination**
 - Observation Techniques
 - Preliminary Orbit Determination
- **Chapter 4: Waltzing Matilda Relative Orbital Motion**
 - Relative Orbital Motion
 - Linearization & State Transition Matrix
- **Chapter 5: To boldly go... Interplanetary Trajectories & Optimization**
 - Lambert's Problem
 - Patched Conics
 - Gravity-assisted fly-by trajectories
- **Chapter 6: Torn between two bodies Restricted Three-Body Problem**
 - N-Body Problem
 - Circular Restricted Three-Body Problem
 - Natural Dynamics around Libration Points
- **Chapter 7: You spin me right round Attitude Dynamics**
 - Attitude representation
 - Attitude Dynamics
- **Chapter 8: In the spin cycle Attitude Motion**
 - Torque-Free Motion
 - Attitude Maneuvers
- **Summary and Revision**

New topic: Spacecraft Attitude



<https://www.youtube.com/watch?v=iwwE-m917CY>



- Week 8: Representation & Kinematics
 - Coordinate systems
 - Rotations, angular velocity
 - Attitude Representation
- Week 9: Dynamics
 - Moment of Inertia, principal axes
 - Constants of motion (angular momentum, kinetic energy)
 - Euler's equations
- Week 10: Torque-free motion & stability
 - Torque free motion
 - Nutation, precession
 - Stability & Damping
- Week 11: Maneuvers & Actuators
 - Coning maneuvers, Yoyo de-spin
 - Gravity gradient
 - Reaction wheels
 - Control moment gyros

References

Orbital Mechanics for Engineering Students

Curtis, H.

Butterworth-Heinemann, 2013.
ISBN-13 978-0123747785

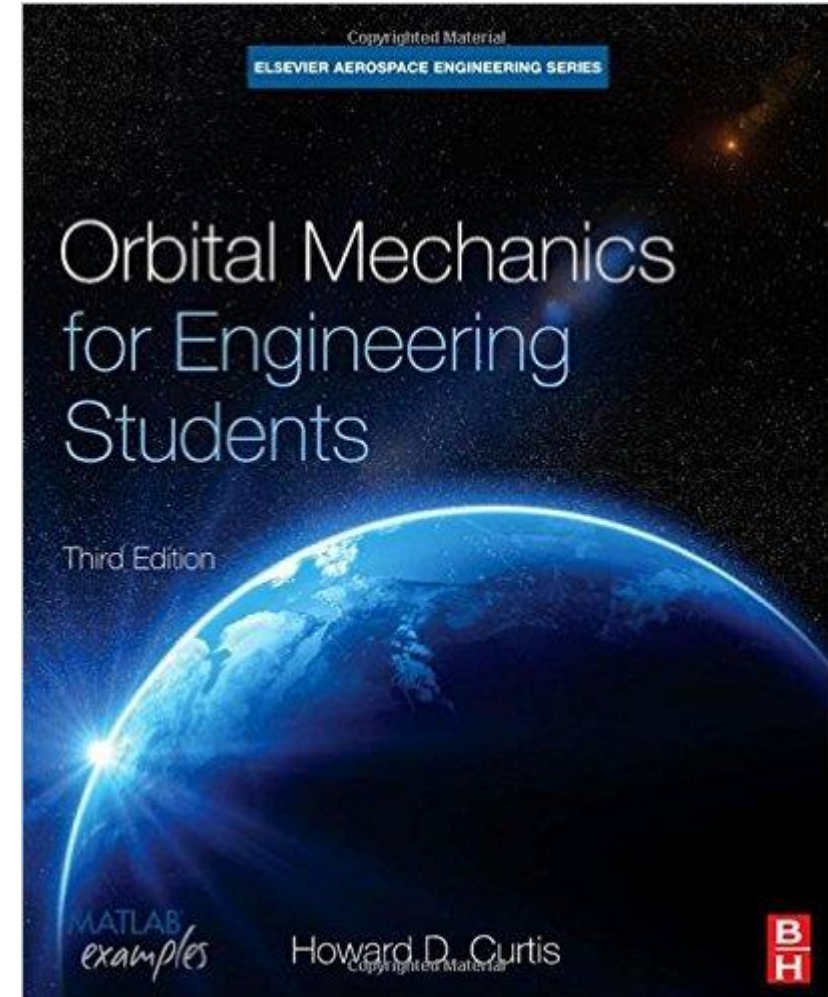
Spacecraft Attitude Dynamics and Control

Chobotov, V.

Krieger Publishing, 1991
ISBN-13 978-0894640315

Subjective observations:

- good introductory textbook
- very explicit mathematical derivations
- copious amount of worked examples
- Limited exposure to control aspects



Orbital Mechanics for Engineering Students

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Spacecraft Attitude Dynamics and Control

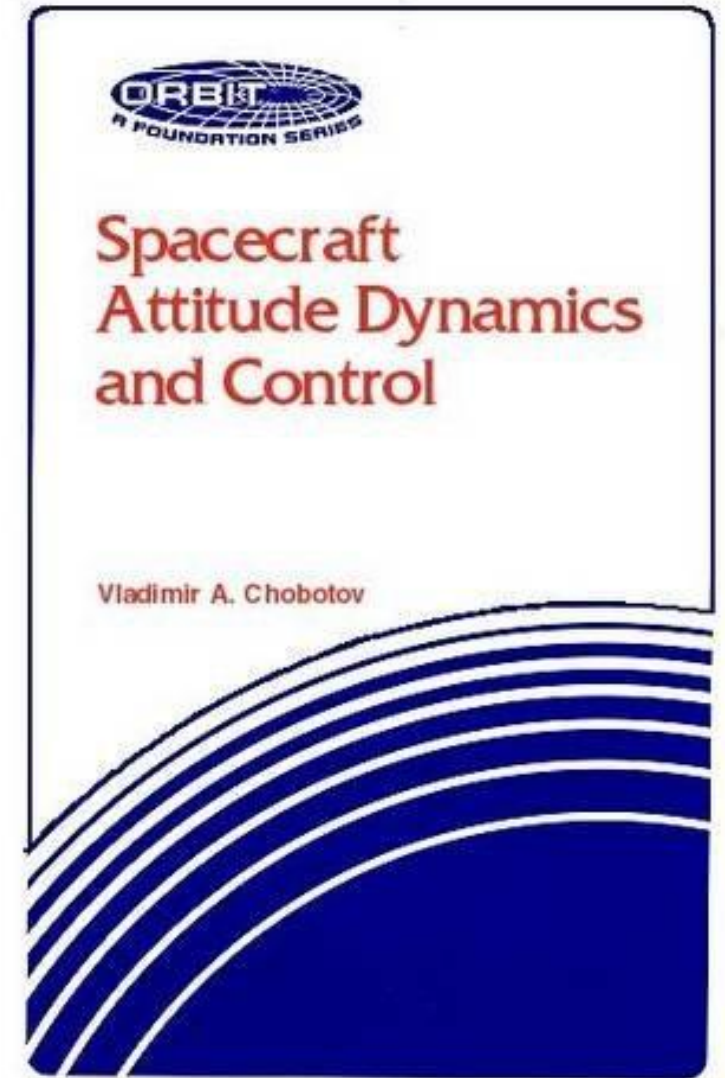
Chobotov, V.

Krieger Publishing, 1991

ISBN-13 978-0894640315

Subjective observations:

- lots of consideration for practical problems
- concise statement of results with references to scientific literature
- few complete derivations
- some practical considerations a bit dated
- Chobotov was expert on control aspects



- Rigid Body Attitude
 - Euler's rotation theorem
 - Angular velocity vector
 - Spacecraft attitude control

- Attitude representation
 - Coordinate systems and vectors
 - Direction Cosines
 - Euler Angles
 - Quaternions

- Kinematics
 - Kinematics of Direction Cosines, Euler Angles, Quaternions

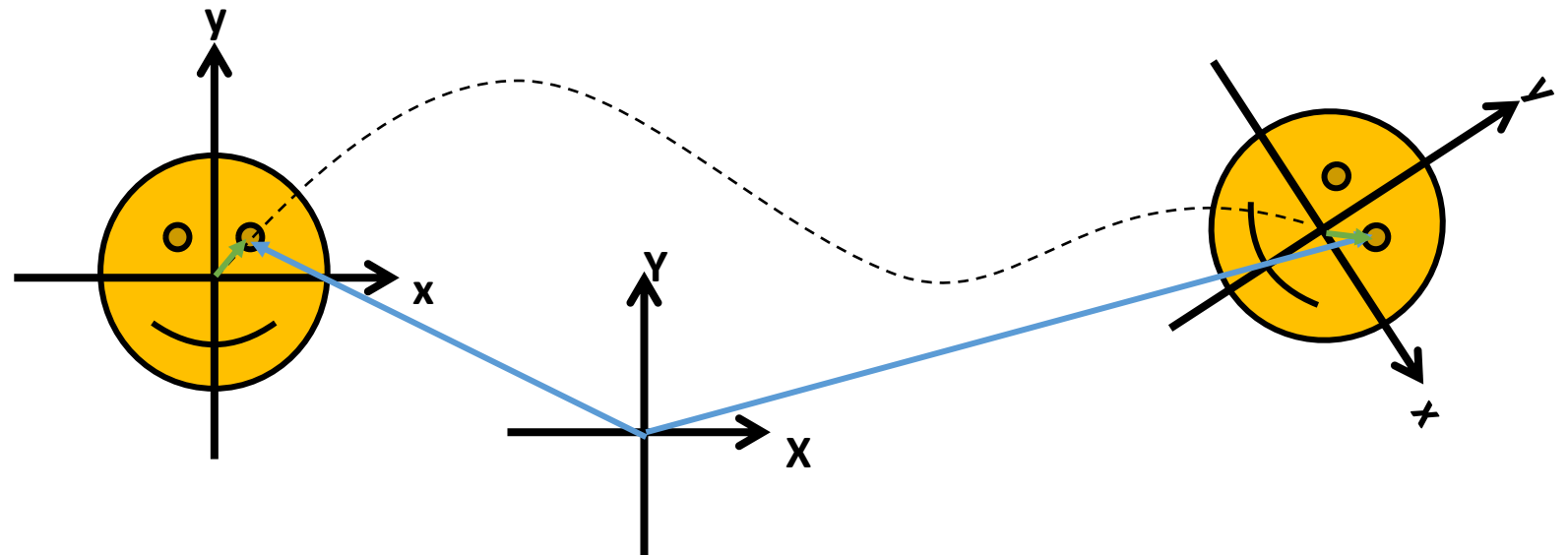
Rigid Body & Spacecraft Attitude



Rigid body

Body in which all distances between its parts remain constant at all times.

Every rigid body admits a **body fixed coordinate system** attached to the body and moving with it.





Rigid body

Body in which all distances between its parts remain constant at all times.

Every rigid body admits a **body fixed coordinate system** attached to the body and moving with it.

- Flexing
- Sloshing
- Fuel expulsion
- Release of components
- Reconfiguration

- Position

Motion: change of position (velocity)

- Orientation

Rotation: change of orientation (angular velocity)

- Position

Motion: change of position (velocity)

- Orientation

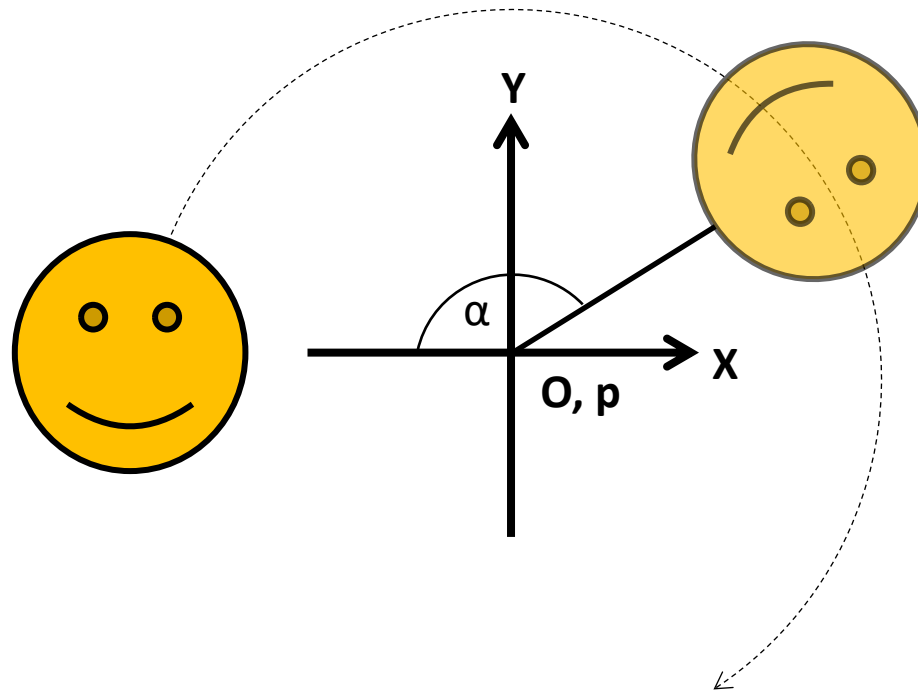
Rotation: change of orientation (angular velocity)

Euler's rotation theorem: In 3D any rigid body state can fully be described by

- displacement of reference point $\mathbf{p}$ (relative to origin $\mathbf{O}$)
- rotation around axis through $\mathbf{p}$ (relative to coordinate axes $\mathbf{X}, \mathbf{Y}, \mathbf{Z}$ at $\mathbf{O}$)

Euler's rotation theorem: In 3D any rigid body state can fully be described by

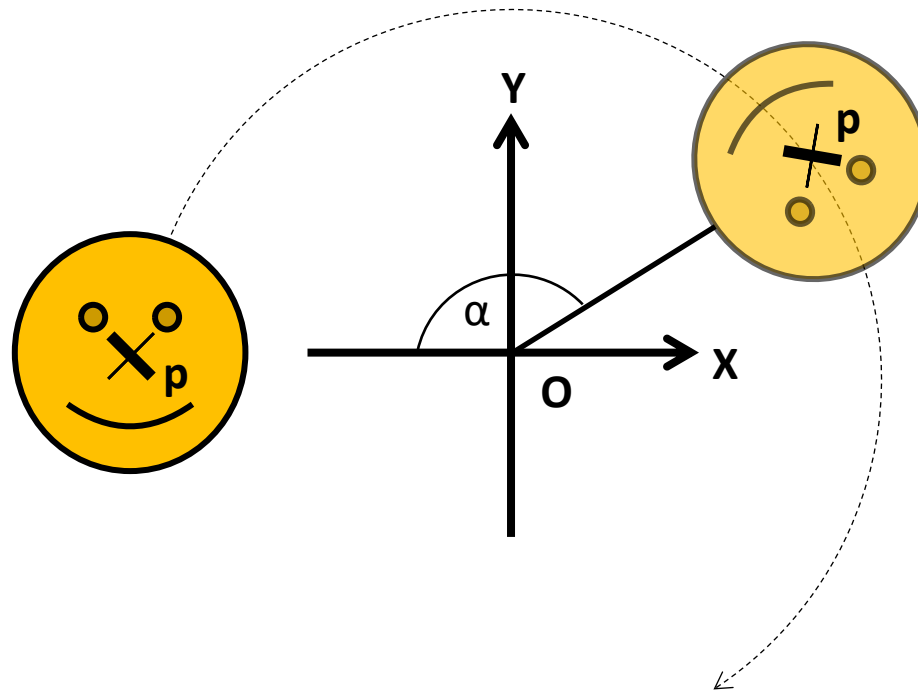
- displacement of reference point $\mathbf{p}$ (relative to origin $\mathbf{O}$)
- rotation around axis through $\mathbf{p}$ (relative to coordinate axes $\mathbf{X}, \mathbf{Y}, \mathbf{Z}$ at $\mathbf{O}$)



$\mathbf{p}$ fixed
+ rotation by α
around $\mathbf{Z}$ -axis

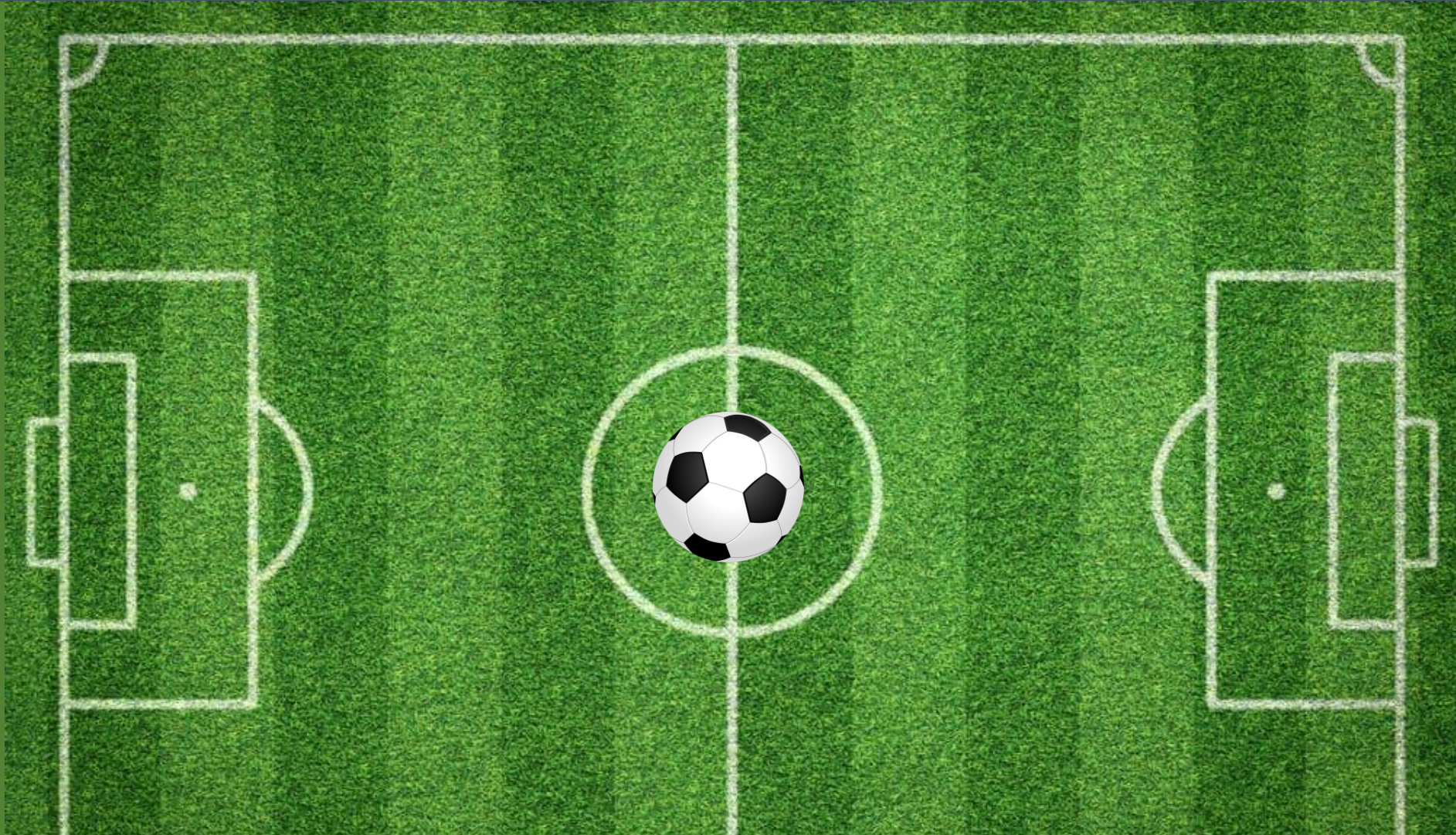
Euler's rotation theorem: In 3D any rigid body state can fully be described by

- displacement of reference point $\mathbf{p}$ (relative to origin $\mathbf{O}$)
- rotation around axis through $\mathbf{p}$ (relative to coordinate axes $\mathbf{X}, \mathbf{Y}, \mathbf{Z}$ at $\mathbf{O}$)



$\mathbf{p}$ moving along circle
+ rotation by α
around $\mathbf{Z}$ -axis

Choice of reference points depends on problem!



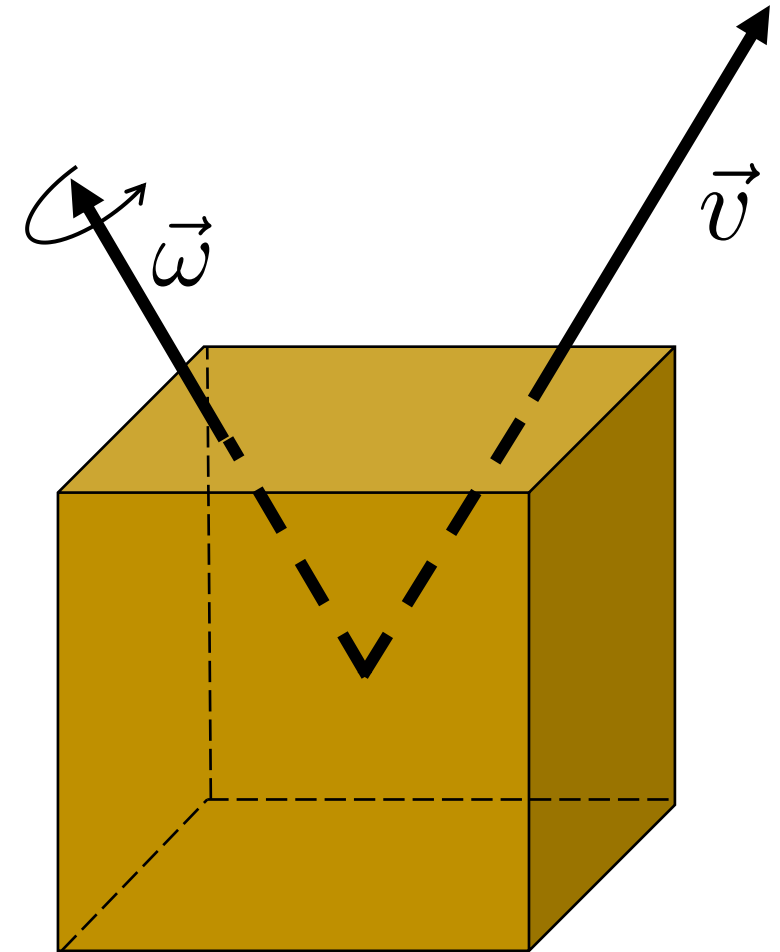
Consequence of **Euler's rotation theorem**: Soccer ball placed at center point at start and end of match always has two points in exact same location

Consequence of Euler's rotation theorem:

Movement of rigid body can be decomposed:

1. linear movement of reference point
2. momentary rotation around axis through reference point

- **Linear** velocity vector $\vec{v}$
- **Angular** velocity vector $\vec{\omega}$
 - direction represents rotation axis
 - magnitude represents rotation speed (rad/sec)
 - rotation of moving frame **e** wrt fixed frame **E**: $\vec{\omega}_{e/E}$

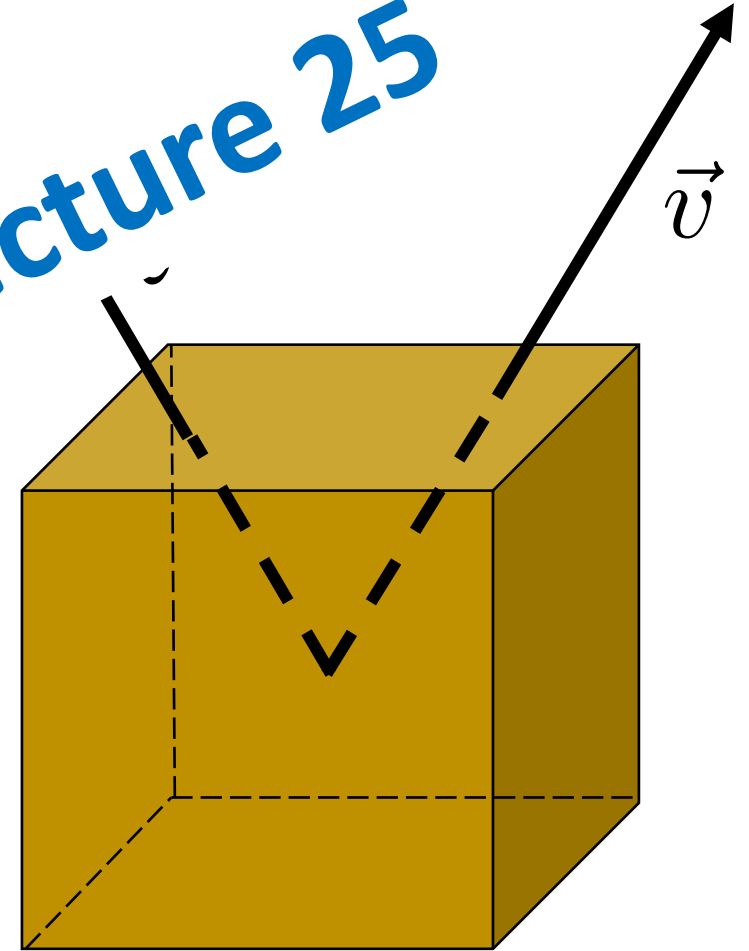


Consequence of Euler's rotation theorem:

Movement of rigid body can be decomposed:

1. linear movement of reference point
2. momentary rotation around axis through reference point

- Linear velocity vector $\vec{v}$
- Angular velocity
 - direction: rotation axis
 - magnitude: rotation speed (rad/sec)
 - rotation of moving frame **e** wrt fixed frame **E**: $\vec{\omega}_{e/E}$



see Chapter 7 Video Lecture 25

- Spacecraft
 - position → Orbital Mechanics
 - orientation → Attitude Dynamics
- Dynamics can couple:
 - position influences orientation
 - orientation influences position
- Objectives
 - Understand evolution of attitude
 - Stabilize attitude
 - Control attitude



(Mostly) passive:

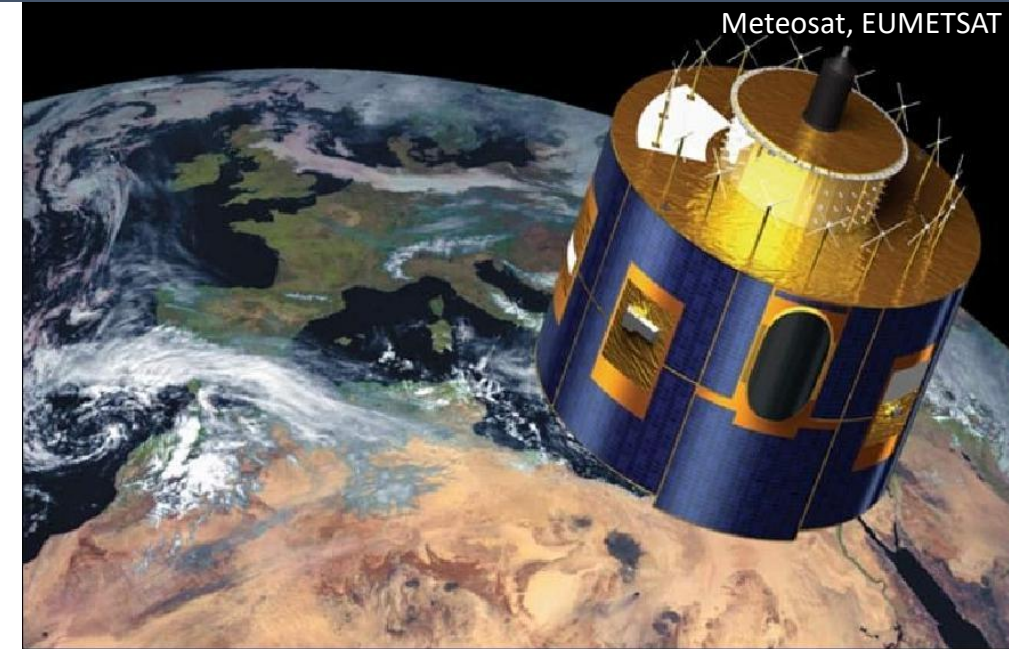
S/C relies on natural dynamics to stabilize

- Spin stabilization
- Gravity gradient
- Magneto-torquers

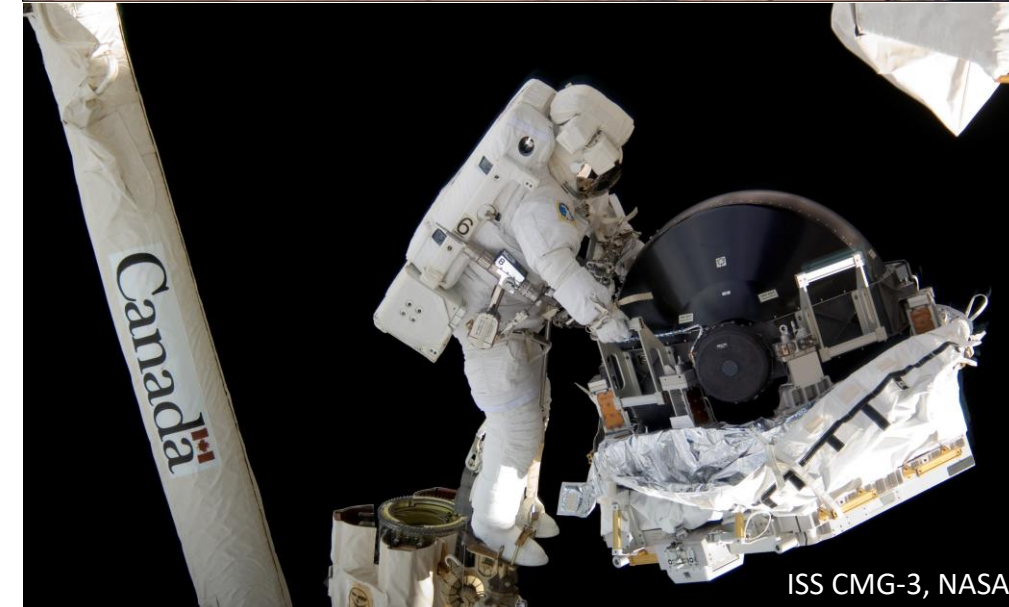
Active (three-axis stabilization):

S/C has active control system

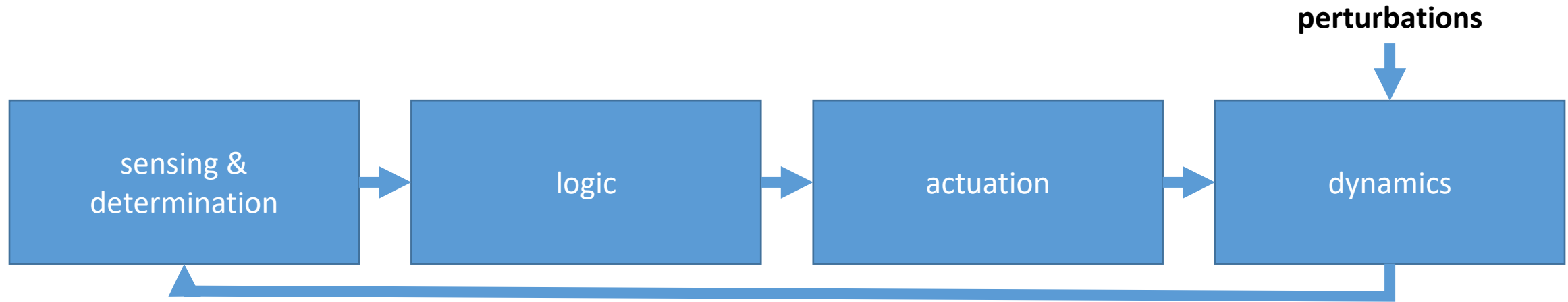
- Attitude thrusters
- Reaction wheels
- Control moment gyros



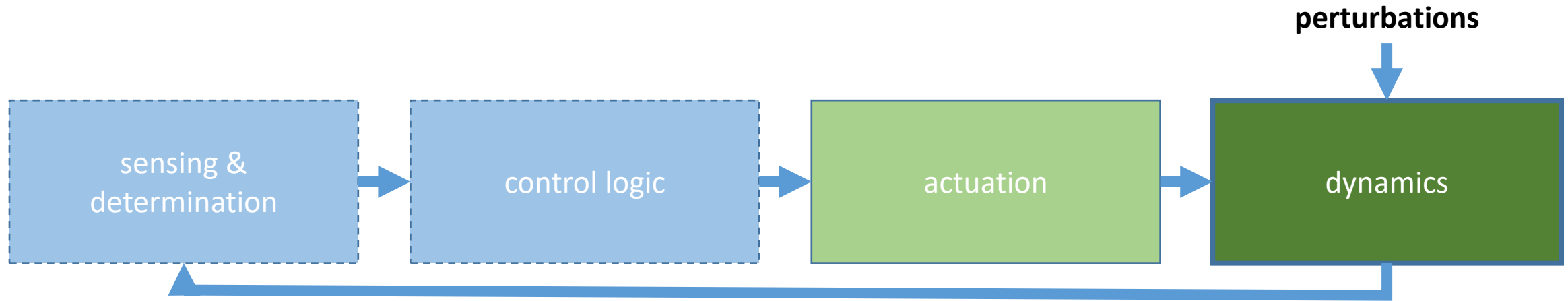
Meteosat, EUMETSAT



ISS CMG-3, NASA



- Combines multiple disciplines
- Control theory to devise logic (e.g. PID, optimal control)
- Closed loop guidance
- On-board or on ground



- Combines multiple disciplines
- Control theory to devise logic (e.g. PID, optimal control)
- Closed loop guidance
- On-board or on ground



Stanford University

Career

- Doctorate on heat conduction in cylinders (University of Hannover, 1929)
- Research engineer on aerodynamics (Aerodynamische Versuchsanstalt Göttingen, 1931)
- Lecturer, Professor on automatic control theory (Stanford University, 1948)

Achievements

- First female full professor at Stanford
- First female Fellow of the AIAA
- Lotz method for wing lift
- Discontinuous and Optimal Control Theory

W.H. Boykin and I. Flügge-Lotz, "*High-accuracy attitude control of satellites in elliptic orbits*," Journal of Optimization Theory and Applications, vol. 5 (1970), pp. 197-224

Attitude Representations

Coordinate system

- Origin + basis
- Basis vectors $\mathbf{e}_i$
 - orthogonal
 - normalized
 - right handed
- Ties abstract math to physical space

Vectors

- Directions
- Expressed w.r.t. a coordinate system

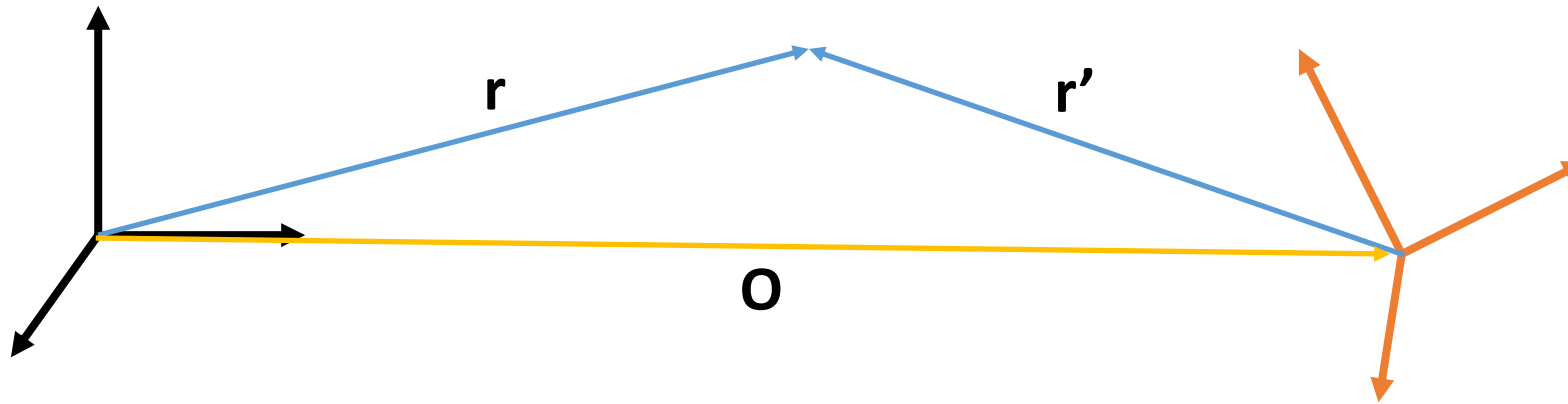
$$\begin{pmatrix} 3 \\ 8 \\ 1 \end{pmatrix}$$

shorthand for

$$3\vec{e}_1 + 8\vec{e}_2 + 1\vec{e}_3$$

Meaningless without coordinate basis

- Same **vector** expressed in different **coordinate systems**



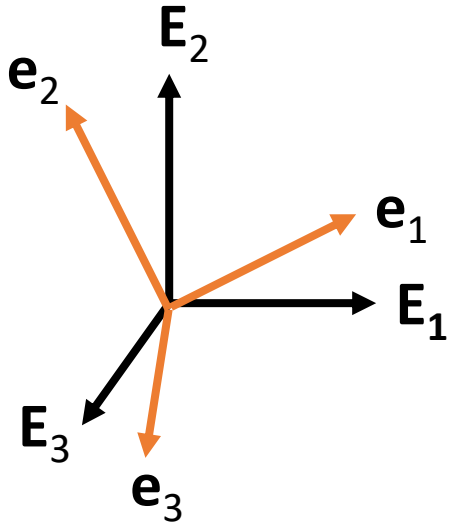
Only options:

- translation of origin $\vec{O} + \vec{r}' = \vec{r}$
- rotation of basis

(note: no explicit basis needed!)

(Euler's theorem!)

Relation between basis vectors $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$ and $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$

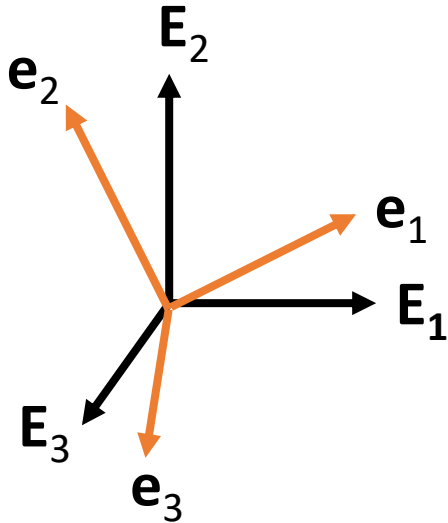


- one basis $\mathbf{E}_i$ at rest (attached to “rest frame”)
- other basis $\mathbf{e}_i$ attached to body (body fixed frame)
- relationship represents orientation of body

Attitude Representations

Direction cosines

Relation between basis vectors $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$ and $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$

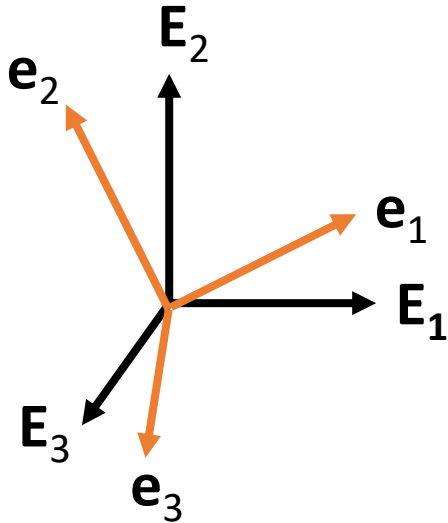


$$\vec{E}_1 = e_{11}\vec{e}_1 + e_{12}\vec{e}_2 + e_{13}\vec{e}_3 = \begin{pmatrix} e_{11} \\ e_{12} \\ e_{13} \end{pmatrix}_e$$

$$\vec{E}_2 = e_{21}\vec{e}_1 + e_{22}\vec{e}_2 + e_{23}\vec{e}_3 = \begin{pmatrix} e_{21} \\ e_{22} \\ e_{23} \end{pmatrix}_e$$

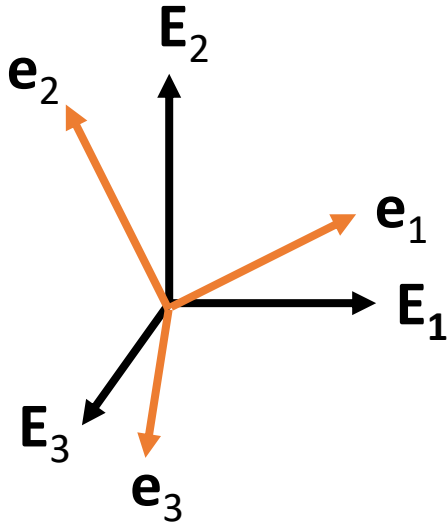
$$\vec{E}_3 = e_{31}\vec{e}_1 + e_{32}\vec{e}_2 + e_{33}\vec{e}_3 = \begin{pmatrix} e_{31} \\ e_{32} \\ e_{33} \end{pmatrix}_e$$

Relation between basis vectors $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$ and $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$



$$\begin{aligned} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}_E &= X_1 \vec{E}_1 + X_2 \vec{E}_2 + X_3 \vec{E}_3 \\ &= \begin{pmatrix} e_{11} & e_{21} & e_{31} \\ e_{12} & e_{22} & e_{32} \\ e_{13} & e_{23} & e_{33} \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}_E \\ &= \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}_e \end{aligned}$$

Relation between basis vectors $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$ and $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$



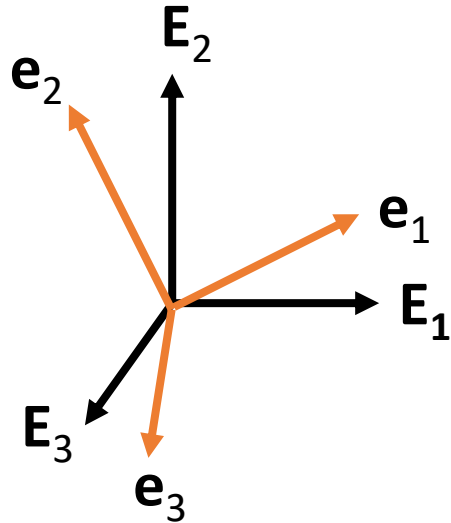
$$R = \begin{pmatrix} e_{11} & e_{21} & e_{31} \\ e_{12} & e_{22} & e_{32} \\ e_{13} & e_{23} & e_{33} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$RR^T = I \Rightarrow R^{-1} = R^T$$

$$(\vec{x})_e = R (\vec{x})_E$$

$$(\vec{x})_E = R^T (\vec{x})_e$$

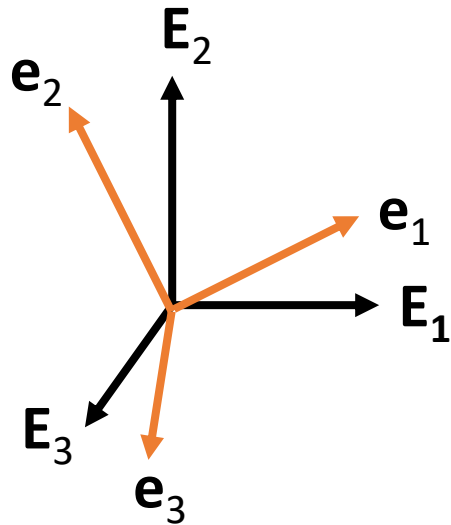
$$e_{ij} = a_{ji} = \vec{E}_i \cdot \vec{e}_j = \cos(\angle \vec{E}_i, \vec{e}_j)$$



Direction cosine matrix

$$R = \begin{pmatrix} e_{11} & e_{21} & e_{31} \\ e_{12} & e_{22} & e_{32} \\ e_{13} & e_{23} & e_{33} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

- represents the relation between rotated coordinate system **e** and fixed system **E**.
- 9 different entries (interdependent!)
 - only 3 independent values
- allow direct transformation of vectors
- **hard to visualize**



Direction cosine matrix

$$R = \begin{pmatrix} e_{11} & e_{21} & e_{31} \\ e_{12} & e_{22} & e_{32} \\ e_{13} & e_{23} & e_{33} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

define relation between rotated
coordinate system e and fixed system E .

9 different entries (interdependent!)

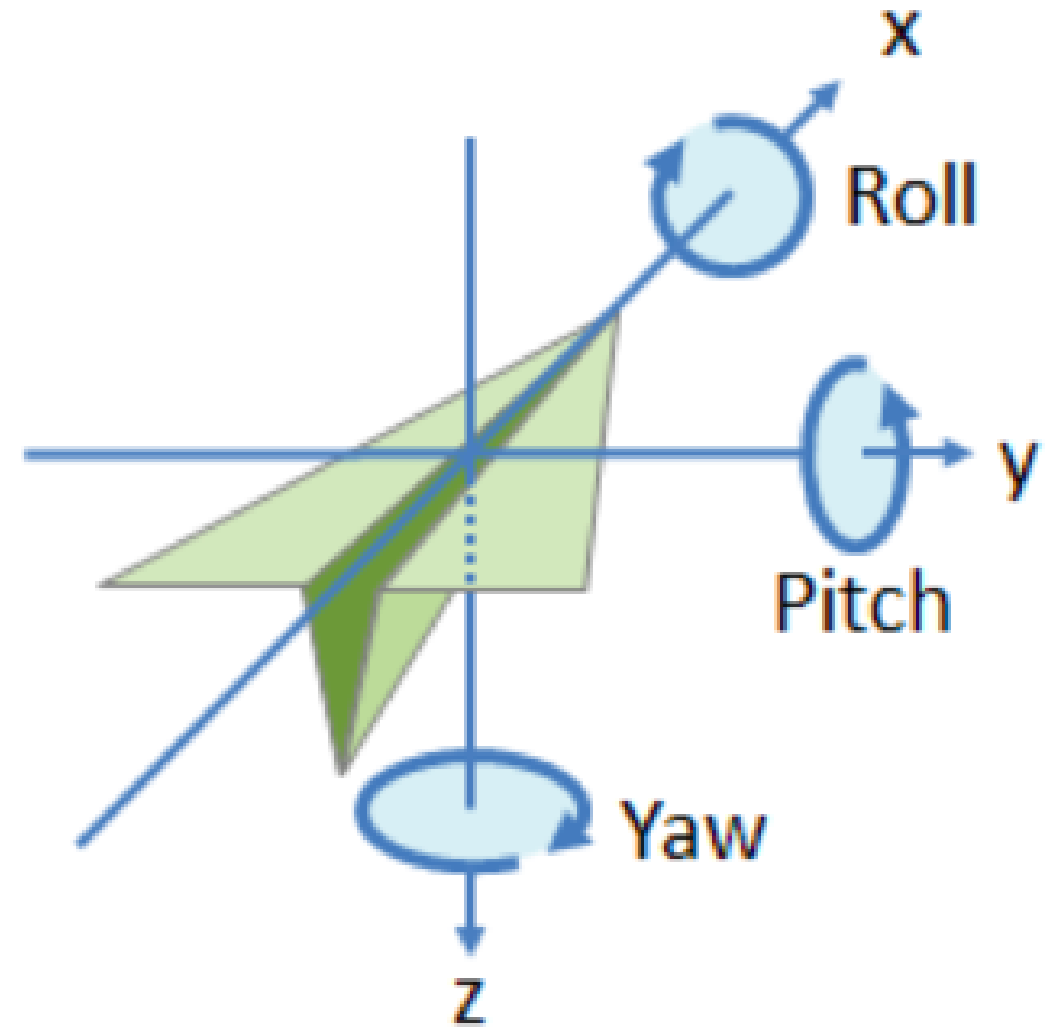
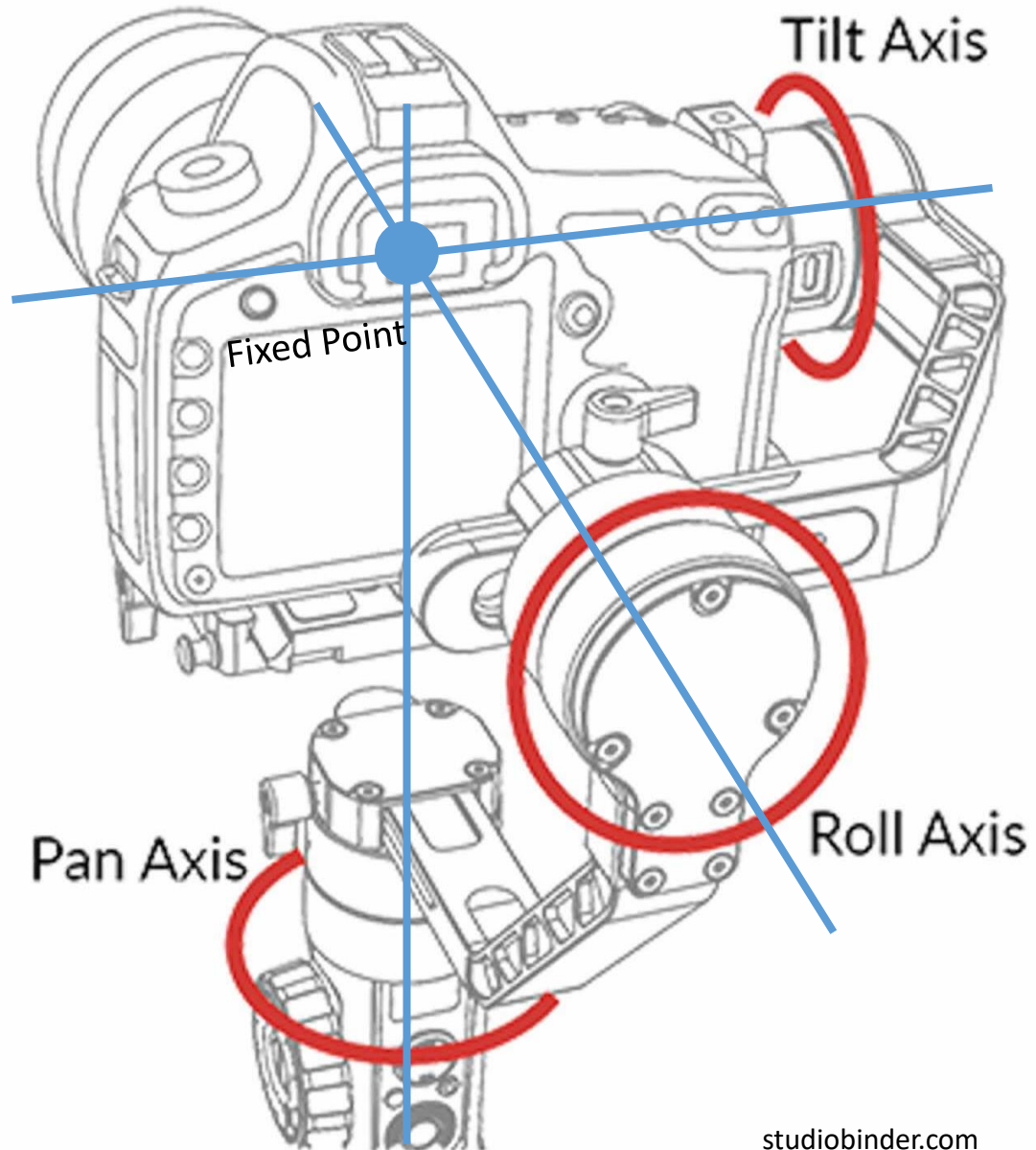
- only 3 independent values
- allow direct transformation of vectors
- **hard to visualize**

see Chapter 7 Video Lecture 26

Attitude Representations

Euler angles

Three Angle Representation



studiobinder.com

Idea

- Only 3 independent quantities
- Intuitive sequence of rotations
- Rotate **E** into **e** frame

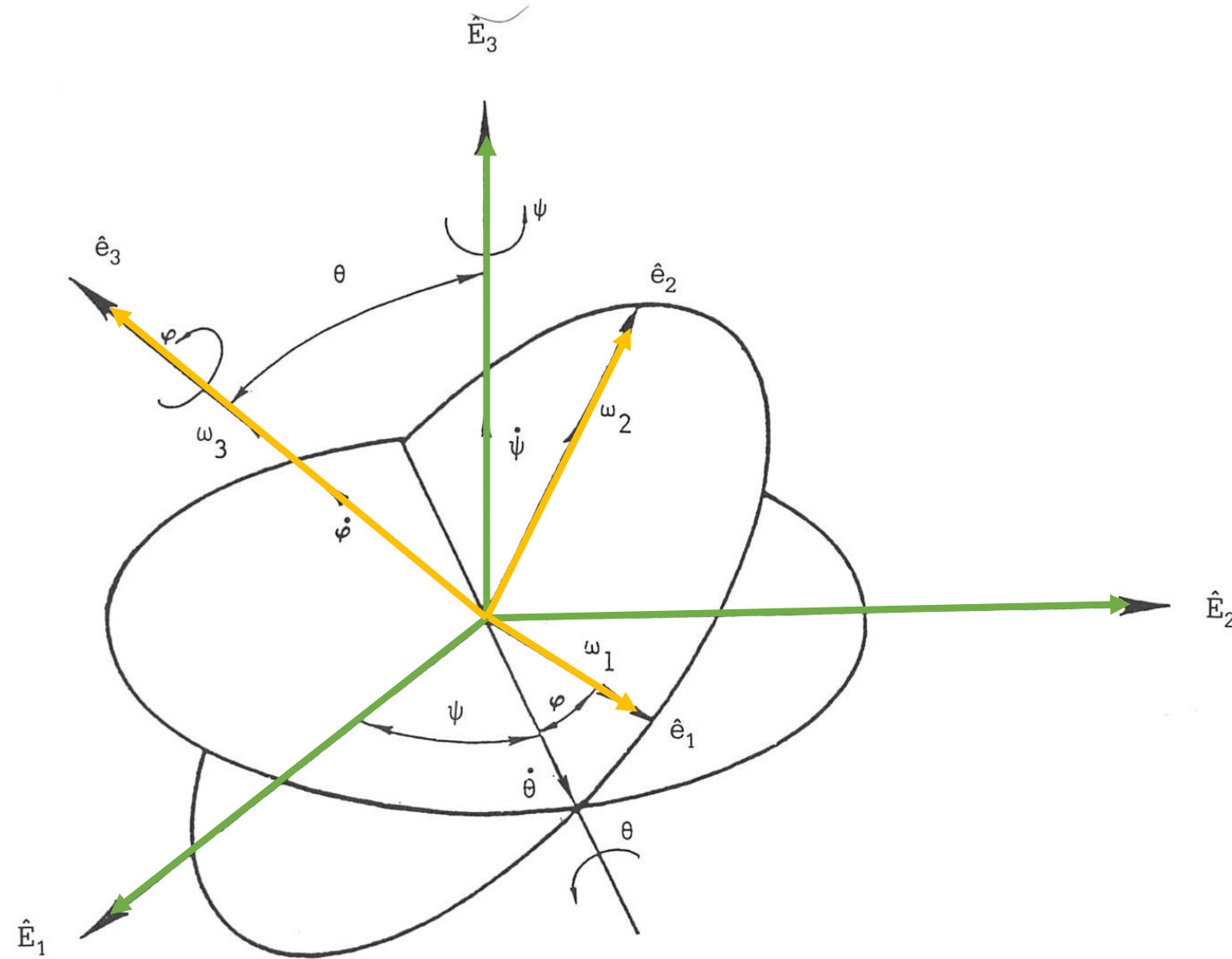


Figure 1.8 Classical Euler rotations of a rigid body.

Chobotov

Idea

- Only 3 independent quantities
- Intuitive sequence of rotations
- Rotate **E** into **e** frame

zxz rotation

1. rotate by ψ around body z-axis
2. rotate by θ around body x-axis
3. rotate by φ around body z-axis

ORDER MATTERS!

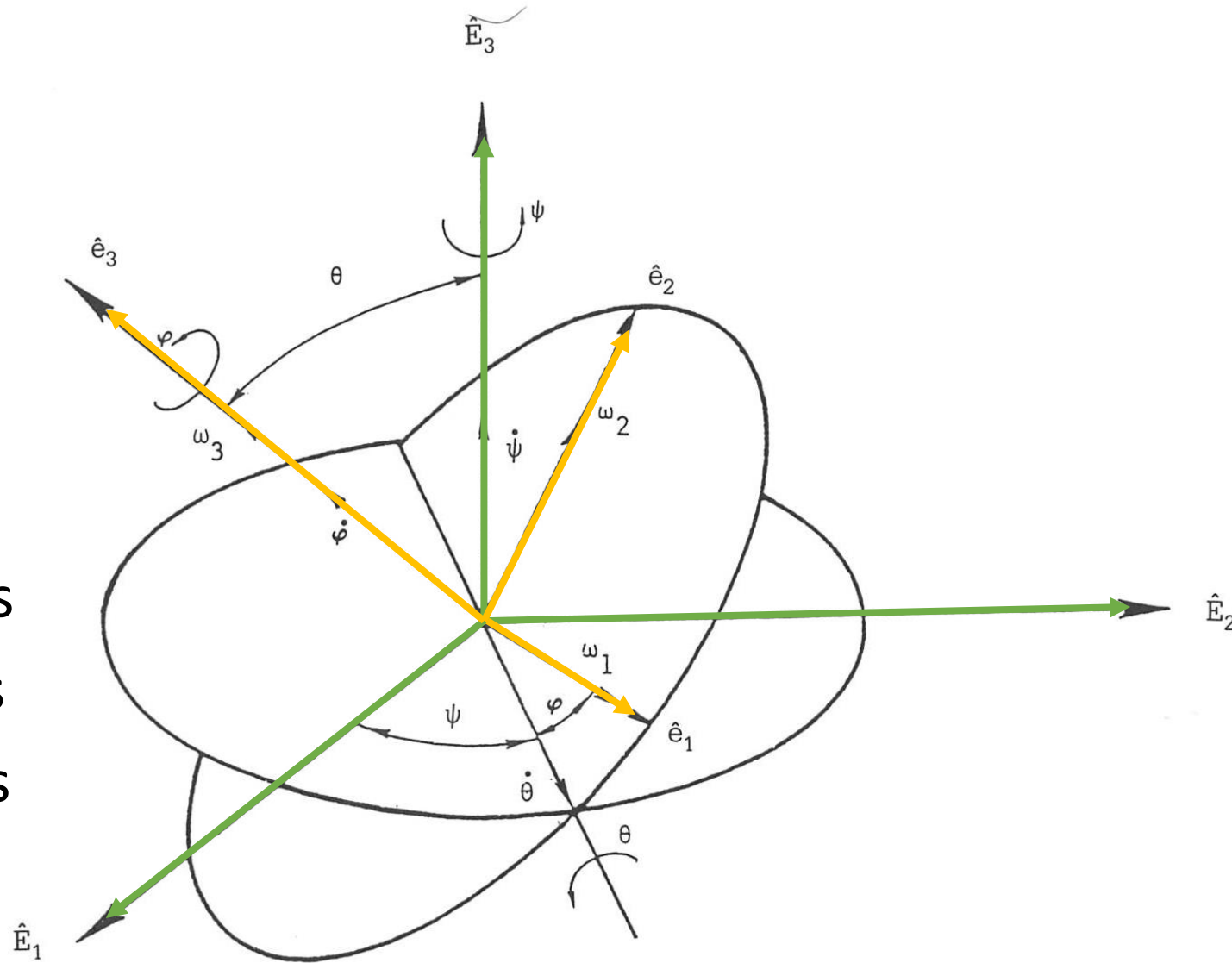


Figure 1.8 Classical Euler rotations of a rigid body.

Chobotov

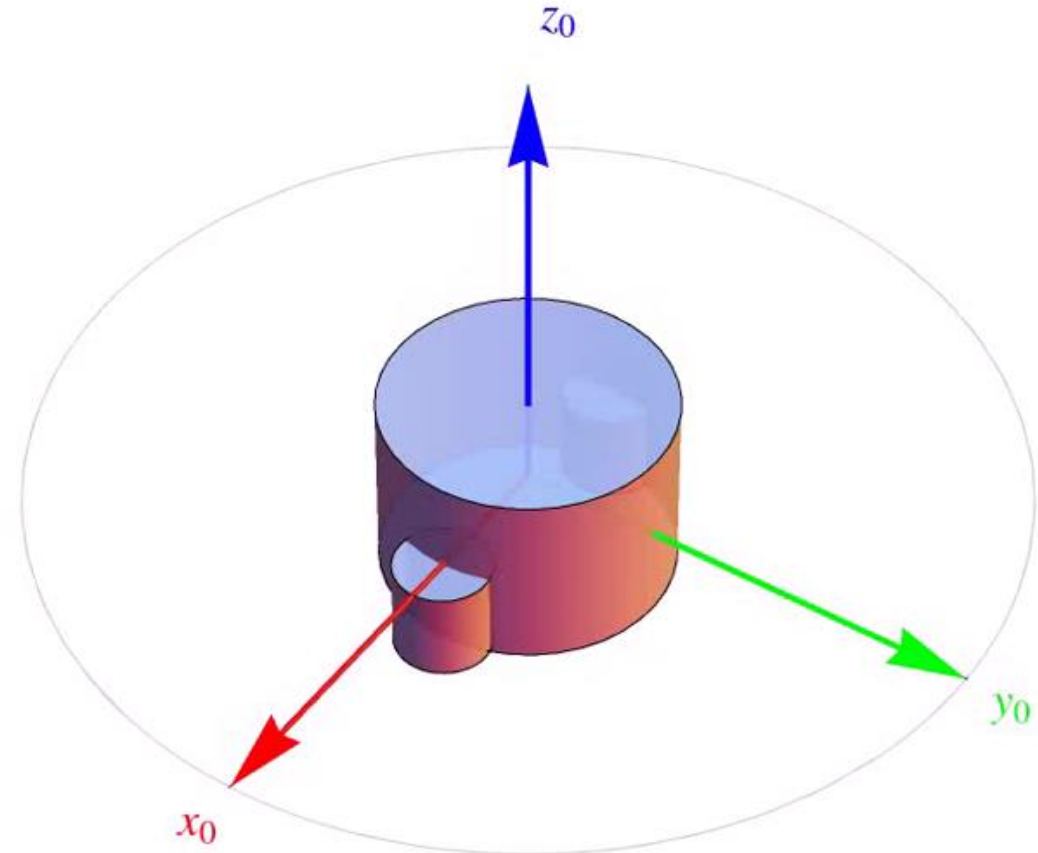
Idea

- Only 3 independent quantities
- Intuitive sequence of rotations
- Rotate **E** into **e** frame

zxz rotation

1. rotate by ψ around body z-axis
2. rotate by θ around body x-axis
3. rotate by φ around body z-axis

ORDER MATTERS!



Animation: Vadim Yudintsev

$$R_{zxz} = \underbrace{\begin{pmatrix} c\phi & s\phi & 0 \\ -s\phi & c\phi & 0 \\ 0 & 0 & 1 \end{pmatrix}}_3 \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & c\theta & s\theta \\ 0 & -s\theta & c\theta \end{pmatrix}}_2 \underbrace{\begin{pmatrix} c\psi & s\psi & 0 \\ -s\psi & c\psi & 0 \\ 0 & 0 & 1 \end{pmatrix}}_1$$

cosine
sine

zxz rotation

1. rotate by ψ around body z-axis
2. rotate by θ around body x-axis
3. rotate by ϕ around body z-axis

$$R_{zxz} = \begin{pmatrix} c\varphi c\psi - s\varphi c\theta s\psi & c\varphi s\psi + s\varphi c\theta c\psi & s\varphi s\theta \\ -s\varphi c\psi - c\varphi c\theta s\psi & -s\varphi s\psi + c\varphi c\theta c\psi & c\varphi s\theta \\ s\theta s\psi & -s\theta c\psi & c\theta \end{pmatrix}$$

direction cosine matrix from Euler angles

zxz rotation

1. rotate by ψ around body z-axis
2. rotate by θ around body x-axis
3. rotate by φ around body z-axis

Euler Angles to Direction Cosines

$$R_{zxz} = \begin{pmatrix} c\varphi c\psi - s\varphi c\theta s\psi & c\varphi s\psi + s\varphi c\theta c\psi & s\varphi s\theta \\ -s\varphi c\psi - c\varphi c\theta s\psi & -s\varphi s\psi + c\varphi c\theta c\psi & c\varphi s\theta \\ s\theta s\psi & -s\theta c\psi & c\theta \end{pmatrix}$$

direction cosine matrix from Euler angles

zxz rotation

1. rotate by ψ around z-axis
2. rotate by θ around body x-axis
3. rotate by φ around body z-axis

see Chapter 7 Video Lecture 27

Attitude Representations

Quaternions

Idea

- Single rotation
 - arbitrary axis
 - arbitrary angle
- Cleverly arranged into 4 (dependent) quantities

$$\vec{m} = \begin{pmatrix} m_1 \\ m_2 \\ m_3 \end{pmatrix}_E$$

rotation axis
unit vector in **E** frame

μ

rotation angle

$$\mathbf{q} = \begin{pmatrix} m_1 \sin(\mu/2) \\ m_2 \sin(\mu/2) \\ m_3 \sin(\mu/2) \\ \cos(\mu/2) \end{pmatrix}$$

Idea

- Single rotation
 - arbitrary axis
 - arbitrary angle
- Cleverly arranged into 4 (dependent) quantities

$$\vec{m} = \begin{pmatrix} m_1 \\ m_2 \\ m_3 \end{pmatrix}_E$$

rotation axis
unit vector in **E** frame

μ

rotation angle

$$q^2 = 1$$

$$\sin(\mu/2) = \sqrt{q_1^2 + q_2^2 + q_3^2}$$

$$\vec{m} = \frac{q_1 \vec{E}_1 + q_2 \vec{E}_2 + q_3 \vec{E}_3}{\sqrt{q_1^2 + q_2^2 + q_3^2}}$$

$$\mathbf{q} = \begin{pmatrix} m_1 \sin(\mu/2) \\ m_2 \sin(\mu/2) \\ m_3 \sin(\mu/2) \\ \cos(\mu/2) \end{pmatrix}$$

Idea

- Single rotation
 - arbitrary axis
 - arbitrary angle
- Cleverly arranged into 4 (dependent) quantities

$$\mathbf{q} = \begin{pmatrix} m_1 \sin(\mu/2) \\ m_2 \sin(\mu/2) \\ m_3 \sin(\mu/2) \\ \cos(\mu/2) \end{pmatrix}$$

$$R = \begin{pmatrix} q_1^2 - q_2^2 - q_3^2 + q_4^2 & 2(q_1 q_2 + q_3 q_4) & 2(q_1 q_3 - q_2 q_4) \\ 2(q_1 q_2 - q_3 q_4) & -q_1^2 + q_2^2 - q_3^2 + q_4^2 & 2(q_1 q_4 + q_2 q_3) \\ 2(q_1 q_3 + q_2 q_4) & 2(-q_1 q_4 + q_2 q_3) & -q_1^2 - q_2^2 + q_3^2 + q_4^2 \end{pmatrix}$$

- No singularities!

Idea

- Single rotation
 - arbitrary axis
 - arbitrary angle
- Cleverly arranged into 4 (dependent) quantities

$$q = \begin{pmatrix} m_1 \sin(\mu/2) \\ m_2 \sin(\mu/2) \\ m_3 \sin(\mu/2) \\ \cos(\mu/2) \end{pmatrix}$$

$$R = \begin{pmatrix} q_1^2 - q_2^2 - q_3^2 + q_4^2 & 2(q_1q_3 - q_2q_4) & 2(q_1q_4 + q_2q_3) & 2(q_2q_3 + q_1q_4) \\ 2(q_1q_2 - q_3q_4) & q_1^2 + q_2^2 - q_3^2 + q_4^2 & 2(-q_1q_4 + q_2q_3) & 2(q_1q_3 - q_2q_4) \\ 2(q_1q_3 + q_2q_4) & 2(q_1q_4 + q_2q_3) & -q_1^2 - q_2^2 + q_3^2 + q_4^2 & 2(q_2q_3 + q_1q_4) \\ 2(q_2q_3 - q_1q_4) & 2(-q_1q_4 + q_2q_3) & 2(q_1q_3 - q_2q_4) & q_1^2 + q_2^2 - q_3^2 + q_4^2 \end{pmatrix}$$

- No singularities!

see Chapter 7 Video Lecture 28

Kinematics

- Linear motion: $\vec{v} = \frac{d}{dt}\vec{x} = \dot{\vec{x}}$
- Velocity as **time derivative of state**
- Given velocity, state can be integrated:

$$\vec{x}(t) = \int_{t_0}^t \vec{v} dt + \vec{x}_0$$

What is “velocity”?

- Linear motion: $\vec{v} = \frac{d}{dt}\vec{x} = \dot{\vec{x}}$
- Velocity as **time derivative of state**
- Given velocity, state can be integrated:

$$\vec{x}(t) = \int_{t_0}^t \vec{v}(t) dt + \vec{x}_0$$

- Same idea for orientation:
e.g. direct cosine matrix
(same for other representations)

$$R = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$v_R = \frac{d}{dt}R = \dot{R}$$

$$= \begin{pmatrix} \dot{a}_{11} & \dot{a}_{12} & \dot{a}_{13} \\ \dot{a}_{21} & \dot{a}_{22} & \dot{a}_{23} \\ \dot{a}_{31} & \dot{a}_{32} & \dot{a}_{33} \end{pmatrix}$$

- Given v_R , R can be integrated



$$R = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$v_R = \frac{d}{dt}R = \dot{R}$$

$$= \begin{pmatrix} \dot{a}_{11} & \dot{a}_{12} & \dot{a}_{13} \\ \dot{a}_{21} & \dot{a}_{22} & \dot{a}_{23} \\ \dot{a}_{31} & \dot{a}_{32} & \dot{a}_{33} \end{pmatrix}$$

- Given v_R , R can be integrated
- Physical meaning of v_R ?
- Must connect to forces, motion, ...
- Derivative not observable!

$$R = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$\begin{aligned} v_R &= \frac{d}{dt} R = \dot{R} \\ &= \begin{pmatrix} \dot{a}_{11} & \dot{a}_{12} & \dot{a}_{13} \\ \dot{a}_{21} & \dot{a}_{22} & \dot{a}_{23} \\ \dot{a}_{31} & \dot{a}_{32} & \dot{a}_{33} \end{pmatrix} \end{aligned}$$

What are Kinematics?

How to get from angular velocity vector to orientation representation?

$$\begin{array}{c}
 R = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \\
 \uparrow \int_{t_0}^t \\
 v_R = \frac{d}{dt} R = \dot{R} \\
 = \begin{pmatrix} \dot{a}_{11} & \dot{a}_{12} & \dot{a}_{13} \\ \dot{a}_{21} & \dot{a}_{22} & \dot{a}_{23} \\ \dot{a}_{31} & \dot{a}_{32} & \dot{a}_{33} \end{pmatrix} \\
 \uparrow ? \\
 \boxed{\vec{\omega}(t)}
 \end{array}$$

What are Kinematics?

Kinematics!

Connect time evolution
of attitude representation with angular velocity

- Similarity to Cartesian linear motion
- Given $\boldsymbol{\omega}(t)$, allows integration of attitude
 - usually no analytical solution
 - uses numerical ODE integration
- Doesn't tell how $\boldsymbol{\omega}(t)$ evolves
 - $\rightarrow$ dynamics
- Following equations are for $\boldsymbol{\omega}$ in **body fixed frame**: $\vec{\omega} = \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}_e$

$$R = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$\int_{t_0}^t$$

$$v_R = \frac{d}{dt} R = \dot{R}$$

$$= \begin{pmatrix} \dot{a}_{11} & \dot{a}_{12} & \dot{a}_{13} \\ \dot{a}_{21} & \dot{a}_{22} & \dot{a}_{23} \\ \dot{a}_{31} & \dot{a}_{32} & \dot{a}_{33} \end{pmatrix}$$

Kinematics

$$\vec{\omega}(t)$$

■ Direction Cosines:

$$\begin{pmatrix} \dot{a}_{11} & \dot{a}_{12} & \dot{a}_{13} \\ \dot{a}_{21} & \dot{a}_{22} & \dot{a}_{23} \\ \dot{a}_{31} & \dot{a}_{32} & \dot{a}_{33} \end{pmatrix} = \begin{pmatrix} \omega_3 a_{21} - \omega_2 a_{31} & \omega_3 a_{22} - \omega_2 a_{32} & \omega_3 a_{23} - \omega_2 a_{33} \\ \omega_1 a_{31} - \omega_3 a_{11} & \omega_1 a_{32} - \omega_3 a_{12} & \omega_1 a_{33} - \omega_3 a_{13} \\ \omega_2 a_{11} - \omega_1 a_{21} & \omega_2 a_{12} - \omega_1 a_{22} & \omega_2 a_{13} - \omega_1 a_{23} \end{pmatrix}$$

■ Euler angles:

$$\begin{pmatrix} \dot{\psi} \\ \dot{\theta} \\ \dot{\varphi} \end{pmatrix} = \frac{1}{\sin \theta} \begin{pmatrix} \sin \varphi & \cos \varphi & 0 \\ \cos \varphi \sin \theta & -\sin \varphi \sin \theta & 0 \\ -\sin \varphi \cos \theta & -\cos \varphi \cos \theta & \sin \theta \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}$$

■ Quaternions:

$$\begin{pmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & \omega_3 & -\omega_2 & \omega_1 \\ -\omega_3 & 0 & \omega_1 & \omega_2 \\ \omega_2 & -\omega_1 & 0 & \omega_3 \\ -\omega_1 & -\omega_2 & -\omega_3 & 0 \end{pmatrix} \begin{pmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{pmatrix}$$

$$\vec{\omega} = \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}_e$$

Kinematic equations

Direction Cosines:

$$\begin{pmatrix} \dot{a}_{11} & \dot{a}_{12} & \dot{a}_{13} \\ \dot{a}_{21} & \dot{a}_{22} & \dot{a}_{23} \\ \dot{a}_{31} & \dot{a}_{32} & \dot{a}_{33} \end{pmatrix} = \begin{pmatrix} \omega_3 a_{21} - \omega_2 a_{31} & \omega_3 a_{22} - \omega_2 a_{32} & \omega_3 a_{23} - \omega_2 a_{33} \\ \omega_1 a_{31} - \omega_3 a_{11} & \omega_1 a_{32} - \omega_3 a_{12} & \omega_1 a_{33} - \omega_3 a_{13} \\ \omega_2 a_{11} - \omega_1 a_{21} & \omega_2 a_{12} - \omega_1 a_{22} & \omega_2 a_{13} - \omega_1 a_{23} \end{pmatrix}$$

Euler angles:

$$\begin{pmatrix} \dot{\psi} \\ \dot{\theta} \\ \dot{\varphi} \end{pmatrix} = \frac{1}{\sin \theta} \begin{pmatrix} \sin \varphi & \cos \varphi & \sin \theta \\ \cos \varphi \sin \theta & -\sin \varphi \sin \theta & \cos \theta \\ -\sin \varphi \cos \theta & \cos \varphi \cos \theta & \sin \theta \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}$$

Quaternions:

$$\begin{pmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{pmatrix} = \begin{pmatrix} 0 & -\omega_2 & \omega_1 & 0 \\ \omega_2 & 0 & \omega_3 & -\omega_1 \\ \omega_3 & -\omega_1 & 0 & \omega_2 \\ -\omega_1 & \omega_2 & \omega_3 & 0 \end{pmatrix} \begin{pmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{pmatrix}$$

$$\vec{\omega} = \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}_e$$

see Chapter 7 Video Lectures 26, 27, 28

- Global and body-fixed coordinate systems, vectors
- Angular velocity vector
- Three representations of orientation
 - Rotation matrix / direction cosines
 - Euler angles
 - Quaternions
- Kinematics for each representation
connect angular velocity vector and time derivative of representation

Further reading:

Curtis: 9.2, 9.9 – 9.11

Chobotov: 1.2

WARNING: Curtis switches the ψ and φ symbols in Euler angles!

Chapter 7: Attitude Representation & Attitude Dynamics – week 1

Dr. Alexander Wittig